Inverse estimation of growth and material properties in growing elastic bodies

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Introduction

Embryonic tissue and organ development are characterised by continuous, spatially non-homogeneous growth. Experimental growth measurements are inaccurate and tedious [1]. Soft living matter can be modelled as an active elastic material with a space-dependent growth strain field [2]. The main challenge is an inverse problem: inferring this hidden growth and potentially non-homogeneous material properties, from measurable displacements. Addressing this ill-posed problem requires optimisation and regularisation to ensure stable, physically meaningful solutions [3].

Methods

The strain tensor is additively decomposed into elastic and growth contributions as under the infinitesimal strain assumption. Mechanical equilibrium in Ω, is solved with Dirichlet and Neumann boundary conditions. The weak form leads to the discrete system Two inverse problems are considered given some

$$\mathbf{K}\mathbf{u} = \mathbf{F}_{\text{ext}} + \mathbf{F}_g$$

displacement data: i) inferring with known Young´s modulus distribution $E(x)$, and ii) inferring $E(x)$ with known . Both are expressed as least-squares minimisations with Tikhonov regularisation, subject to equilibrium. Synthetic displacement data are generated using FEM with 100 Q4 elements, with fixed bottom corners and top traction for case 2. Regularisation parameter β is fitted from errors in displacement and material.

i) Inferring growth strain with known

$$\min_{\mathbf{u},\boldsymbol{\epsilon}_g} \mathcal{J}(\mathbf{u},\boldsymbol{\epsilon}_g) = \frac{1}{2}\|\mathbf{u}-\mathbf{u}_{\text{syn}}\|_2^2 + \frac{\beta}{2}\|\boldsymbol{\epsilon}_g\|_2^2$$

$$\text{s.t.} \quad \mathbf{K}\mathbf{u} - \mathbf{G}\boldsymbol{\epsilon}_g - \mathbf{F}_{ext} = 0$$

ii) Inferring with known

$$\min_{\mathbf{u},\mathbf{E}} J(\mathbf{u},\mathbf{E}) = \frac{1}{2}\,|\,\mathbf{u}-\mathbf{u}_{\text{syn}}\|_2^2 + \frac{\beta}{2E_{\text{ref}}}\,|\,\mathbf{E}-\mathbf{E}_{\text{ref}}\|_2^2$$

$$\text{s.t.} \quad \mathbf{K}(\mathbf{E})\mathbf{u} - \mathbf{G}(\mathbf{E})\boldsymbol{\epsilon}_g - \mathbf{F}_{\text{ext}} = 0$$

Optimisation conditions are followed through a Lagrangian, leading to a KKT system

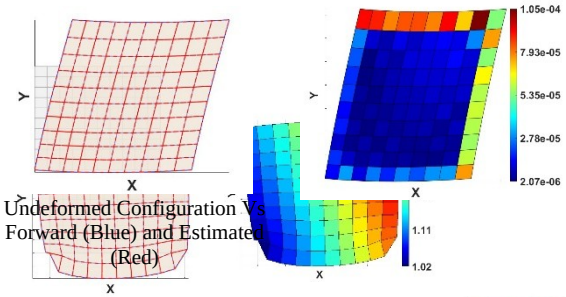
$$(\mathbf{u}-\mathbf{u}_{\text{syn}}) + \mathbf{K}^T\boldsymbol{\lambda} = \mathbf{0},$$
$$\beta\boldsymbol{\varepsilon}_g - \mathbf{G}^T\boldsymbol{\lambda} = \mathbf{0},$$
$$\mathbf{K}\mathbf{u} - \mathbf{G}\boldsymbol{\varepsilon}_g = \mathbf{F}_{\text{ext}}$$

For case 2, the control equation looks like:

$$\beta(\mathbf{E}-\mathbf{E}_{\text{ref}}) + \mathbf{H}^T(\mathbf{u},\mathbf{E})\boldsymbol{\lambda} = \mathbf{0}$$

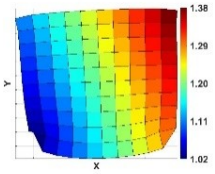
Results

Case 1:



Undeformed Configuration Vs Forward (Blue) and Estimated (Red)

Case 2:



Undeformed Configuration Forward (Blue) and Estimated (Red)

Conclusion

Case 1: At $\beta=10^{-6}$: The KKT optimisation yields an accurate reconstruction displacement (8.44E-6 relative error), while growth is recovered at (5.54E-5) from the available displacement data.

Case 2: At $\beta=10^{-8}$: The inversion achieves accurate displacement reconstruction (7.329E-4 relative error), while Young's modulus is recovered at (1.391E-2 relative error). This limitation stems from the severe ill-conditioning of matrix (κ=4.984E14) and its non-trivial null space (nullity 92), leading to weak identifiability of the Young's modulus distribution.

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References

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